

Supplementary Information for

Physics-guided discriminability of process-level observables in partially observable cancer systems

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Overview

This Supplementary Information provides conceptual and interpretative support for the main manuscript. While the Supplementary Methods describe the data sources, preprocessing procedures, gene-set mapping, patient-level observable construction, survival analysis, random-baseline generation, λ -scan analysis, coherence filtering, and K-scale observation-lens analysis, the present document focuses on how the resulting measurements should be interpreted.

The central aim of the manuscript is not to reconstruct a unique latent cancer state from bulk transcriptomic data. Instead, the study develops an operational framework for evaluating process-level transcriptomic observables under partial observability. In this setting, bulk gene-expression profiles are treated as projected, heterogeneous, and biologically constrained observations of cancer systems. Biological-process and pathway-level observables are therefore interpreted as effective coarse-grained measurements rather than direct readouts of complete underlying biological states.

The Supplementary Notes address six interpretative issues. First, we explain why the patient-level observable width $W_{B,p}$, defined from the within-process gene-expression distribution of a biological-process observable B in patient p , can be interpreted as an operational proxy for observable uncertainty. Second, we clarify why survival was used as an operational validation endpoint rather than as a direct representation of latent-state truth. Third, we discuss why pathway-level or biological-process-level averaging remains useful despite the loss of gene-level detail. Fourth, we explain why high-discriminability random gene sets do not invalidate the biological interpretation of structured observables, but instead

motivate random-background and coherence-based diagnostics. Fifth, we discuss scale sensitivity, including the distinction between the number of genes within an observable, N_B , and the number of selected observables used in patient-level scoring, K . Finally, we define the boundaries of inference from projected transcriptomic observables and clarify what the present framework does and does not claim.

Together, these notes support the main interpretation that observable width does not universally improve clinical discriminability. Rather, width changes how discriminability should be interpreted. Activity magnitude, observable compactness, random-background behavior, biological structure, and observation scale should be considered jointly when evaluating the usefulness of process-level transcriptomic observables.

Supplementary Note S1

Why patient-level observable width $W_{B,p}$ is interpreted as an operational uncertainty proxy

Bulk gene-expression profiles provide high-dimensional molecular observations of tumor samples, but they do not directly reveal a unique underlying biological state. Each bulk measurement reflects a mixture of malignant cells, immune components, stromal cells, vascular structures, technical variation, and other hidden tissue-level or molecular factors. Under this condition, a biological-process or pathway-level observable should not be interpreted as a direct scalar measurement of a single biological mechanism. Instead, it is more appropriately treated as an effective projected observation constructed from a set of genes associated with a biological process.

For a biological-process observable B in patient p , the present study represents the observable using two complementary quantities. The first is the mean process activity, $\mu_{B,p}$, which summarizes the average direction and magnitude of

expression change among genes belonging to B . The second is the observable width, $W_{B,p}$, which summarizes the internal dispersion of gene-expression values within the same process-level observable in the same patient. Importantly, $W_{B,p}$ is defined at the **patient level** and within the genes of a given biological process. It is not the dispersion of pathway scores across patients.

This distinction is central to the interpretation of the study. A process-level observable with a high mean activity but narrow internal width represents a relatively coherent expression pattern among its member genes in that patient. In contrast, a process-level observable with similar mean activity but broad internal width indicates that the genes assigned to the same process show more heterogeneous expression behavior. Such heterogeneity may arise from biological diversity within the process, mixed cellular composition, partial activation of submodules, opposing regulation among member genes, pathway crosstalk, or measurement noise. Therefore, the same average process activity may correspond to different internal observable structures.

The use of FWHM to quantify $W_{B,p}$ is intended as an operational summary of this internal width. FWHM provides a simple and interpretable measure of the spread of the within-process gene-expression distribution. It is less dependent on extreme values than a full range and is easier to interpret than more abstract distributional metrics. In this study, FWHM is not used to claim that the within-process expression profile follows a specific parametric distribution. Rather, it is used as a practical descriptor of how broad or compact the observed expression cloud is within a biological-process gene set.

The term “observable uncertainty proxy” should therefore be interpreted carefully. $W_{B,p}$ is not an exact entropy, not a full posterior uncertainty estimate, and not a direct measurement of the true latent-state distribution. Instead, it is an operational indicator of observable dispersion within a constrained biological-process projection. A broader $W_{B,p}$ suggests that the process-level observable is less internally compact in that patient, whereas a narrower $W_{B,p}$ suggests that

the member genes behave more coherently under the chosen observation mapping.

This interpretation is especially relevant under partial observability. When the underlying cancer system cannot be fully reconstructed from available measurements, observable structure becomes important. Two patients may show similar mean pathway activity but differ in the internal width of that pathway-level observable. Conversely, two biological processes may show similar average activity across a cohort but differ substantially in their width distributions. These differences may affect how confidently a process-level observable can be interpreted, selected, or used for clinical stratification.

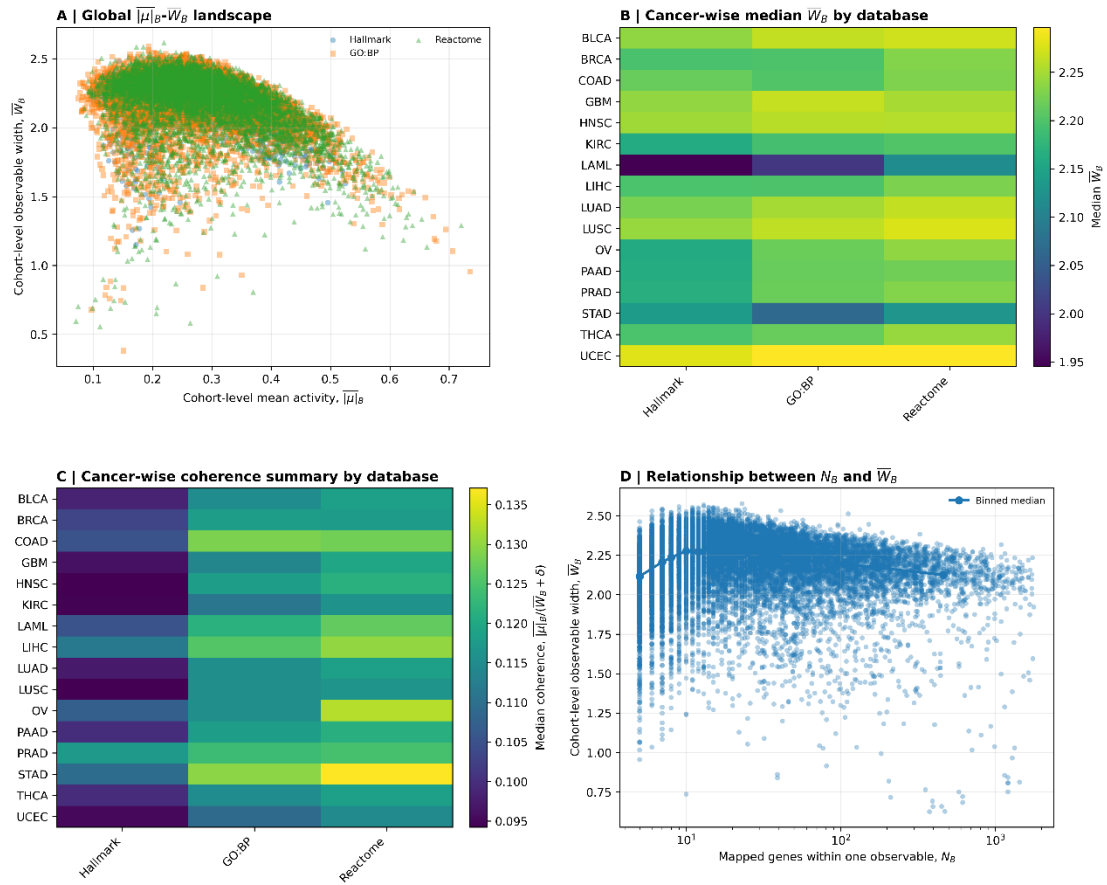
The activity–width representation therefore extends conventional pathway-level scoring by separating two aspects of the observable: the location of the process-level signal and the thickness of the process-level expression cloud. The mean activity $\mu_{B,p}$ captures the central tendency of the observable, whereas $W_{B,p}$ captures its internal dispersion. This does not imply that width alone determines biological relevance or clinical utility. Instead, width provides an additional diagnostic dimension that should be interpreted together with activity magnitude, survival-based discriminability, random-background behavior, coherence, and biological annotation.

In this sense, $W_{B,p}$ is best understood as a minimal operational proxy for uncertainty in process-level transcriptomic observation. It does not replace mechanistic modeling, single-cell resolution, spatial profiling, or causal perturbation experiments. However, it provides a practical way to quantify how compact or dispersed a process-level observable appears within each patient using standard bulk gene-expression data. This makes it useful for evaluating whether a pathway or biological-process score represents a coherent observable or whether the same scalar score may conceal substantial internal heterogeneity. The main contribution of this width-aware formulation is therefore not that $W_{B,p}$ universally improves discriminability. Rather, it changes the interpretation of

discriminability. A high-discriminability observable with compact width may be interpreted differently from a high-discriminability observable with broad internal dispersion. Similarly, a low-discriminability observable may still be biologically interpretable but poorly aligned with the clinical endpoint being tested. By introducing $W_{B,p}$, the present framework provides an additional layer for distinguishing activity magnitude, observable compactness, and outcome-oriented discriminability under partial observability.

The cross-cancer activity-width structure of the resulting process-level observables is summarized in Supplementary Figure S1, which shows that observable width varies across cancer cohorts, gene-set databases, and gene-set aggregation scales.

Supplementary Figure S1. Complete real-data activity-width structure across cancers



N_B denotes mapped genes within one BP/pathway observable; K is not used in this figure.

Supplementary Figure S1. Complete real-data activity–width structure across cancers.

(A) Global activity–width landscape of process-level transcriptomic observables across cancer cohorts and gene-set databases. Each point represents a biological-process or pathway observable summarized by cohort-level mean absolute activity, $|\bar{\mu}|_B$, and cohort-level observable width, $\bar{W}_B$. (B) Cancer-wise median observable width by gene-set database, showing database- and cancer-dependent differences in process-level observable dispersion. (C) Cancer-wise median coherence summary, computed as $|\bar{\mu}|_B/(\bar{W}_B + 1)$, providing a descriptive measure of activity relative to observable width. (D) Relationship between mapped gene-set size, N_B , and cohort-level observable width. The binned median trend indicates that observable width depends on the internal aggregation scope of each biological-process or pathway observable. Here, N_B denotes the number of mapped genes within one observable, whereas K , used in the observation-lens analysis, denotes the number of selected process-level observables included in a patient-level score.

Supplementary Note S2

Why survival was used as an operational endpoint under partial observability

The present study uses survival-based stratification as an operational endpoint for evaluating process-level transcriptomic observables. This choice should be interpreted carefully. Survival is not treated as a direct readout of the latent cancer state, nor as a complete representation of disease progression. Instead, it is used as a clinically meaningful endpoint through which the usefulness of an observable can be tested under partial observability.

In an ideal setting, one would evaluate whether a molecular observable accurately tracks the latent state of the cancer system over time. For example, an ideal validation design would measure whether an observable can distinguish whether a patient's system state moves toward a more malignant, resistant,

unstable, or treatment-insensitive region after a defined interval or intervention. Such validation would require longitudinal, spatially resolved, and clinically annotated measurements of tumor state, treatment exposure, microenvironmental composition, molecular adaptation, and outcome trajectories. These data are rarely available at sufficient scale across many cancer cohorts.

Bulk transcriptomic datasets, especially large public cancer cohorts, usually provide cross-sectional molecular measurements together with clinical endpoints such as overall survival or progression-related outcomes. Under these conditions, survival provides a practical operational endpoint for testing whether a process-level observable is aligned with clinically relevant variation. If an observable separates patients into groups with different survival outcomes, this suggests that the observable captures some information related to clinically meaningful system differences. However, this does not mean that the observable directly reconstructs the hidden biological state responsible for the survival difference.

This distinction is important. Under partial observability, the true latent state of a cancer system is not fully accessible from bulk gene-expression data. The observed transcriptomic profile is a projected measurement generated through tissue composition, gene regulation, pathway activity, tumor heterogeneity, microenvironmental interaction, technical variation, and other hidden factors. Therefore, the relationship between an observable and survival is indirect. A survival-discriminative observable may reflect one or more latent biological factors, including proliferation, immune suppression, stromal remodeling, hypoxia, metabolic adaptation, treatment sensitivity, residual disease burden, or other clinically relevant processes. Conversely, a biologically meaningful observable may fail to separate survival if it is not aligned with the endpoint, if the endpoint is too coarse, or if clinical follow-up and event numbers are insufficient.

For this reason, survival-based discriminability is used in this study as a task-

specific operational criterion rather than as a universal measure of biological truth. The survival-based discriminability score evaluates whether a process-level observable can separate patients with different clinical outcomes under a specified endpoint and stratification rule. It does not imply that high-discriminability observables are necessarily causal drivers of survival, nor that low-discriminability observables are biologically irrelevant. Instead, it quantifies endpoint alignment: whether the observable contains information useful for distinguishing outcome groups in the available cohort.

This endpoint-based interpretation is especially important when comparing different cancer types. Some cancers may show strong survival separation for particular process-level observables, while others may show weak or inconsistent separation. These differences may arise from biological heterogeneity, treatment context, sample size, censoring structure, event frequency, or endpoint suitability. For example, overall survival may be informative in one cancer type but less sensitive in another if the follow-up duration is short, the number of death events is low, or the disease course is strongly influenced by treatment after sampling. Therefore, survival-based discriminability should be interpreted as cohort- and endpoint-dependent.

Survival was selected because it provides a clinically interpretable and widely available outcome for cross-cancer analysis. It allows process-level observables to be evaluated using a common operational framework across multiple datasets. This is useful for assessing whether activity-width representations, uncertainty-aware scoring, random-background diagnostics, and K-scale observation-lens analyses produce clinically relevant separation. However, survival is only one possible endpoint. Other endpoints, such as progression-free interval, recurrence, treatment response, pathological complete response, time to relapse, or longitudinal molecular transition, may be more appropriate depending on the clinical question and data availability.

The use of survival also provides a conservative test of observable usefulness.

Since survival is influenced by many factors beyond the sampled tumor transcriptome, including treatment, comorbidities, disease stage, residual disease, immune status, and clinical management, a transcriptomic observable that consistently separates survival may capture robust endpoint-relevant information. At the same time, survival is a coarse endpoint. Failure to separate survival should not be interpreted as proof that an observable lacks biological meaning. It may simply indicate that the observable is not strongly aligned with the survival endpoint used in the analysis.

Within the framework of this study, survival-based discriminability therefore serves three roles. First, it provides an operational validation endpoint for testing whether process-level observables are clinically informative. Second, it allows comparison between activity-only and width-aware representations. Third, it provides a reference task for evaluating random-background behavior, coherence filtering, and observation-scale sensitivity.

The main claim is not that survival reveals the true latent cancer state. Rather, survival is used because it provides an available and clinically meaningful endpoint through which projected transcriptomic observables can be evaluated. Under partial observability, this distinction is essential. The study does not claim to reconstruct latent-state dynamics directly. It instead asks whether process-level observables, when represented by activity and width, provide useful endpoint-aligned information despite the limitations of projected bulk transcriptomic data.

In future studies, the same activity–width and discriminability framework could be applied to more direct dynamic endpoints, such as treatment response, molecular progression, recurrence, spatial transition, or longitudinal state change. Such endpoints would provide a closer link to latent-state movement than cross-sectional survival analysis. In the present work, however, survival provides a practical and clinically interpretable operational endpoint for testing the usefulness and limitations of process-level transcriptomic observables under

partial observability.

Supplementary Note S3

Why pathway-level averaging remains biologically useful under partial observability

Pathway-level and biological-process-level averaging is a coarse-grained representation of transcriptomic data. It reduces many gene-expression measurements into a smaller number of process-level observables. At first glance, this simplification may appear to discard important gene-level information. However, under partial observability, coarse-grained pathway-level averaging remains biologically useful because it provides an interpretable effective projection of high-dimensional molecular variation.

Bulk transcriptomic profiles are generated from heterogeneous tumor tissues. A measured gene-expression value may reflect malignant cells, immune cells, stromal cells, endothelial components, tissue composition, regulatory state, technical variation, and other hidden factors. Therefore, individual gene-level measurements are not direct readouts of isolated biological mechanisms. They are projected observations shaped by multiple biological and technical sources. Under this condition, retaining all gene-level measurements does not automatically solve the problem of partial observability. More features may increase resolution, but they may also increase noise, redundancy, and interpretative ambiguity.

Biological-process and pathway-level averaging provides a practical way to organize this high-dimensional projected information. Instead of treating each gene independently, genes are grouped according to prior biological knowledge, such as shared pathway membership, functional annotation, regulatory program, or biological process. The resulting pathway-level score is not a perfect measurement of the pathway. Rather, it is an effective observable summarizing

the collective expression behavior of genes associated with that biological process.

This coarse-grained representation has several advantages. First, it improves biological interpretability. A single gene may have context-dependent roles and may participate in multiple processes, whereas a pathway-level observable provides a more interpretable link to broader biological programs such as proliferation, immune activation, hypoxia, metabolism, extracellular matrix remodeling, epithelial–mesenchymal transition, DNA repair, or inflammatory signaling. Second, pathway-level averaging can reduce gene-level noise by summarizing coordinated expression patterns across multiple genes. Third, it allows cancer transcriptomes to be compared at the level of biological processes rather than at the level of thousands of individual features.

The purpose of pathway-level averaging in this study is not to claim that all genes within a pathway behave identically. Nor does it assume that a pathway is fully captured by the arithmetic mean of its member genes. Instead, the pathway-level mean activity $\mu_{B,p}$ is treated as a minimal effective activity descriptor. It captures the central tendency of the observed gene-expression values within biological process B for patient p . This central tendency is useful because many biological programs are not driven by a single gene but by coordinated shifts across multiple components.

However, averaging alone is incomplete. A pathway-level mean may conceal substantial internal heterogeneity. Two patients may have the same $\mu_{B,p}$ but very different internal expression patterns among the genes in B . One patient may show a compact, coordinated process-level pattern, while another may show mixed activation, partial suppression, or dispersed expression among the same gene set. This is why the present study pairs pathway-level averaging with patient-level observable width $W_{B,p}$. The mean activity summarizes the location of the process-level signal, while the width summarizes the internal dispersion of the observable.

In this sense, the activity–width representation preserves the usefulness of pathway-level averaging while making its limitations explicit. The average remains biologically useful as a coarse-grained activity measure, but its interpretation is conditioned by the width of the underlying gene-expression distribution. A high average activity with narrow width suggests a more coherent process-level observable. A high average activity with broad width suggests that the same scalar score may conceal heterogeneous gene-level behavior. Therefore, the framework does not reject pathway averaging. It refines its interpretation under partial observability.

This view is consistent with the idea of effective observation. In complex biological systems, complete microscopic reconstruction is often impossible. Instead, useful observables are frequently coarse-grained quantities that summarize relevant collective behavior. In cancer transcriptomics, pathway-level scores serve this role by mapping high-dimensional gene-expression profiles into biologically interpretable process-level coordinates. These coordinates do not fully reconstruct the latent cancer state, but they can help evaluate whether certain biological programs align with outcome-relevant variation.

The random-background analysis further clarifies the role of pathway-level structure. If pathway-level averaging were entirely arbitrary, structured biological-process observables would have little reason to behave differently from size-matched random gene sets. However, the comparison between structured and random observables allows the analysis to test whether biological annotation provides additional interpretability or discriminability beyond random aggregation. Importantly, high-discriminability random gene sets may still occur, so biological annotation alone is not sufficient. This is why pathway-level observables must be evaluated using survival-based discriminability, random-background thresholds, coherence, and biological-overlap diagnostics.

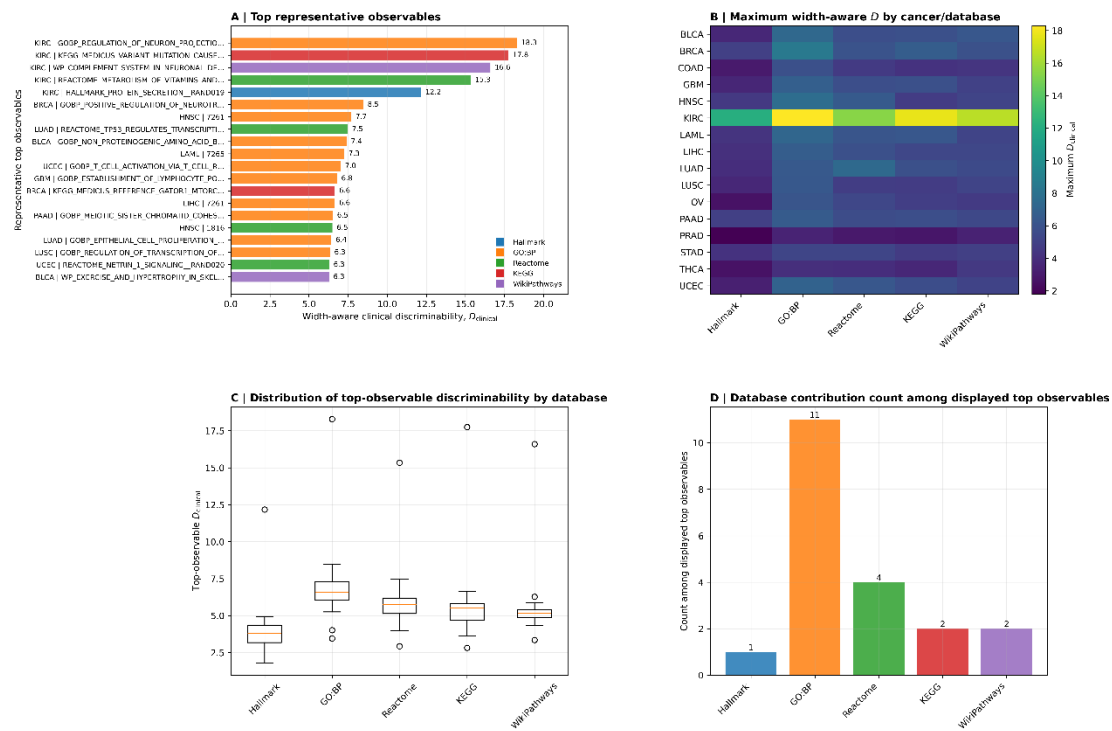
Pathway-level averaging should therefore be understood as useful but not

definitive. It provides an interpretable projection of transcriptomic variation, not a complete mechanistic model. It helps reduce dimensionality, organize gene-expression data according to prior biological knowledge, and generate process-level observables that can be evaluated for clinical discriminability. At the same time, it may miss subpathway structure, gene-specific directionality, network topology, cell-type-specific regulation, spatial heterogeneity, and causal interactions.

The present framework addresses this limitation by avoiding overinterpretation of pathway averages. A pathway-level observable is not considered clinically useful simply because it has a recognizable biological name. Instead, its usefulness depends on whether it shows endpoint-relevant discriminability, whether its behavior exceeds random-background expectation, whether it is internally coherent, and whether its activity-width structure is interpretable within the biological and clinical context.

Thus, pathway-level averaging remains biologically useful under partial observability because it provides a practical coarse-grained observation layer between raw gene-expression measurements and inaccessible latent biological states. It does not eliminate uncertainty, but it organizes uncertainty into interpretable process-level observables. When paired with patient-level width, random-background diagnostics, and survival-based evaluation, pathway-level averaging becomes not merely a dimensionality-reduction step, but a structured way to evaluate which projected transcriptomic observables are informative, coherent, and clinically aligned.

Supplementary Figure S5. Representative top-performing process-level observables across cancer cohorts



Representative top observables were summarized by width-aware discriminability, with supplementary comparison across cancers and gene-set databases.

Supplementary Figure S5. Representative top-performing process-level observables across cancer cohorts.

(A) Top-ranked representative process-level observables across cancer/database pairs according to width-aware clinical discriminability. Only the highest-ranked observables are shown to provide a readable summary of candidate observables with strong endpoint-aligned separation. (B) Maximum width-aware clinical discriminability by cancer cohort and gene-set database. (C) Distribution of top-observable clinical discriminability across gene-set databases. (D) Number of displayed top-ranked observables contributed by each database. These results provide supplementary evidence that high-discriminability process-level observables are distributed across cancer contexts and gene-set databases. The database contribution count should be interpreted descriptively, because it depends on the size, annotation scope, and composition of each gene-set universe rather than indicating that one database is universally superior.

Supplementary Note S4

Why high-discriminability random observables do not invalidate biological structure

Random gene-set analysis was included in this study as a diagnostic control rather than as a competing biological model. Its purpose is to test whether the discriminability observed in structured biological-process or pathway observables clearly exceeds what may arise from size-matched random aggregation of genes. This distinction is important because high survival-based discriminability can occasionally occur even in randomly assembled gene sets. Such cases do not invalidate biological structure. Instead, they show why discriminability must be interpreted against an appropriate random background.

A process-level observable may achieve high clinical discriminability for several reasons. It may represent a biologically meaningful program aligned with disease outcome, such as proliferation, immune suppression, stromal remodeling, hypoxia, or treatment resistance. However, high discriminability may also arise from statistical fluctuation, cohort-specific structure, hidden sample imbalance, correlation with broad transcriptomic axes, or accidental inclusion of genes related to outcome-relevant biology. Therefore, a high value of survival-based discriminability alone is insufficient for biological interpretation.

The random-background analysis addresses this issue by asking a stricter question: does a structured observable perform better than random gene sets of comparable size? Size matching is important because gene-set size can influence the stability and behavior of aggregated observables. A large gene set may average out noise differently from a small gene set, while a small gene set may be more sensitive to individual high-impact genes. By comparing structured observables with size-matched random sets, the analysis evaluates whether biological-process annotations provide discriminability beyond what might be expected from random gene aggregation under similar dimensional conditions.

The observation that some random gene sets can reach high discriminability

should therefore not be viewed as a failure of the biological-process framework. Instead, it provides an important warning: survival-based discriminability is not automatically equivalent to biological interpretability. A random set with high D may not represent a coherent pathway, even if it separates survival groups. Conversely, a structured biological-process observable may be biologically meaningful but show weak survival discriminability if it is not aligned with the endpoint being tested. This is why the present study separates three concepts: biological annotation, observable coherence, and clinical discriminability.

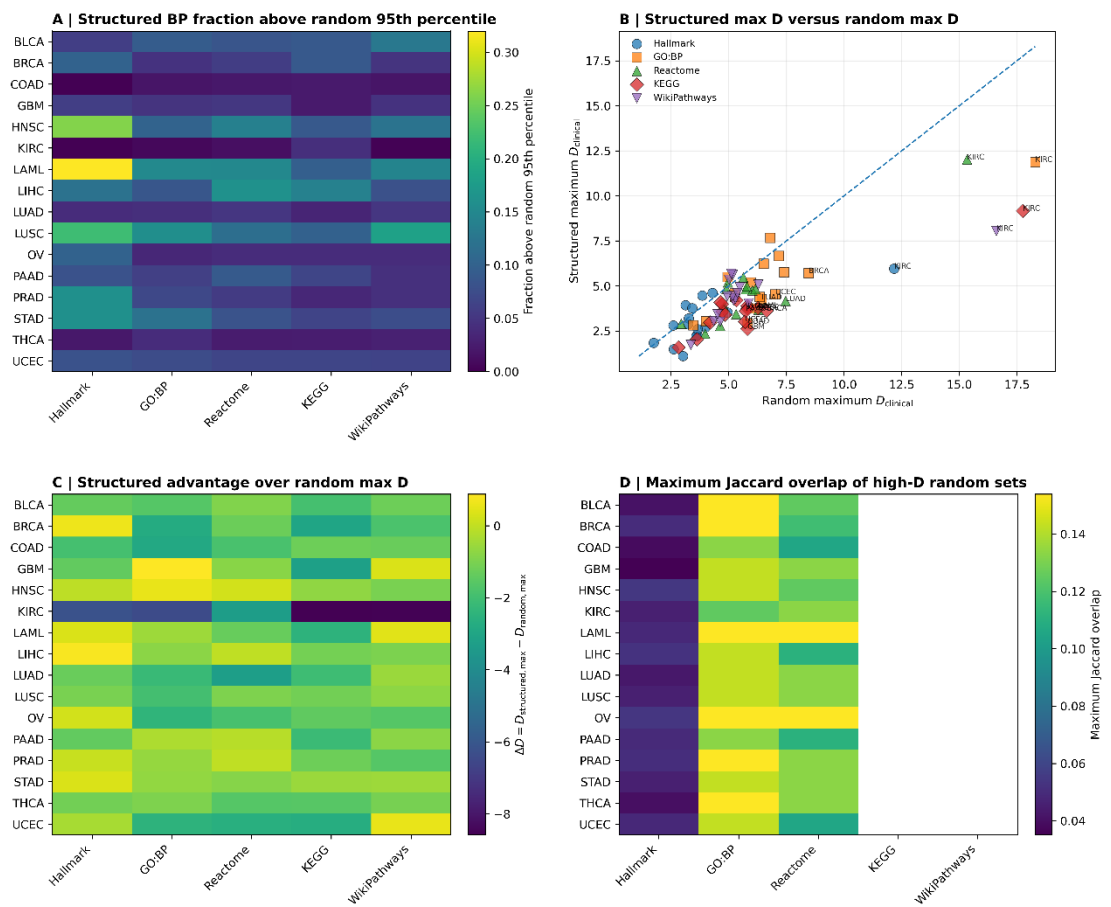
High-D random observables may arise in at least two ways. First, they may occur by chance, especially when many random sets are tested. In this case, they represent a statistical background against which structured observables should be compared. Second, some high-D random sets may contain partial biological structure. Even though the set was generated randomly, it may include genes that overlap with known biological programs or lie near outcome-relevant transcriptomic axes. In bulk cancer transcriptomes, many genes are correlated through shared cell-type composition, proliferation programs, immune infiltration, stromal content, tumor purity, and other broad biological factors. A random gene set may therefore accidentally capture part of a clinically relevant signal.

For this reason, high-D random sets should not be dismissed automatically as meaningless, but they also should not be interpreted as validated biological mechanisms. They are best treated as diagnostic objects. If high-D random sets show little overlap with known annotations and poor coherence, they likely represent statistical or cohort-specific effects. If they show partial overlap with structured annotations, they may indicate that outcome-relevant information is distributed across the transcriptome and not fully captured by curated pathway boundaries. Both possibilities are informative.

This is why the main manuscript uses a diagnostic workflow rather than relying on D alone. Structured observables are evaluated in relation to random-background thresholds, width-aware coherence, and biological-overlap analysis. The random

95th percentile provides a practical reference for asking whether a structured observable exceeds random expectation. The $D +$ coherence filter asks whether an observable is not only clinically discriminative but also internally compact or interpretable in activity–width space. The biological-overlap check, such as maximum Jaccard overlap between high- D random sets and structured annotations, asks whether high-performing random sets may contain partial biological signals.

Supplementary Figure S4. Random-background diagnostics and biological-overlap analysis across cancers



Structured observables were evaluated against random-background expectation, structured-versus-random maximum discriminability, and biological-overlap diagnostics.

Supplementary Figure S4. Random-background diagnostics and biological-overlap analysis across cancers.

(A) Fraction of structured biological-process or pathway observables exceeding the 95th percentile of size-matched random gene-set backgrounds. (B) Comparison between maximum clinical discriminability from structured

observables and maximum clinical discriminability from random gene sets. The dashed diagonal indicates equal maximum discriminability between structured and random observables. (C) Structured advantage over random maximum discriminability, defined as $\Delta D = D_{\text{structured,max}} - D_{\text{random,max}}$. Positive values indicate cancer/database settings in which structured observables achieved stronger maximum discriminability than random gene sets, whereas negative values indicate settings where random gene sets reached comparable or stronger maximum discriminability. (D) Maximum Jaccard overlap between high- D random gene sets and structured biological annotations. This overlap analysis was used as a diagnostic check to assess whether high-performing random gene sets partially intersect known biological-process or pathway structures. Together, these analyses show that high discriminability should be interpreted against random-background expectation and biological-overlap diagnostics rather than from D alone.

The full random-background diagnostic results are summarized in Supplementary Figure S4. These analyses show that structured observables sometimes exceed random-background expectations, but high- D random observables can also occur, supporting the need to interpret discriminability together with random-background behavior and biological-overlap diagnostics.

Together, these diagnostic layers help distinguish several cases. A structured observable with high D , high coherence, and performance above the random 95th percentile is a stronger candidate for biological and clinical interpretation. A structured observable with high D but low coherence should be interpreted cautiously, because its survival separation may be driven by heterogeneous or unstable internal gene behavior. A random observable with high D but low biological overlap may represent background fluctuation or cohort-specific structure. A random observable with high D and substantial overlap with structured annotations may suggest that clinically relevant biology is distributed beyond the exact boundaries of curated gene sets.

Therefore, the presence of high- D random observables strengthens rather than weakens the logic of the study. It demonstrates that clinical discriminability must be handled carefully. Without a random background, one might overinterpret any high- D pathway as biologically meaningful. With a random background, high- D structured observables can be evaluated more rigorously. The random test therefore acts as a safeguard against overclaiming.

This interpretation is also consistent with partial observability. Bulk gene-expression measurements are projected observations of heterogeneous cancer systems. Outcome-relevant information may not be confined to cleanly annotated pathways. It may be distributed across many genes, partially reflected in multiple biological processes, or mixed with tissue composition and microenvironmental signals. As a result, random aggregation can occasionally capture clinically relevant variation, especially when the random set overlaps with broad biological axes. This does not mean that biological structure is irrelevant. It means that biological structure should be evaluated jointly with discriminability and observable coherence rather than assumed from annotation alone.

The role of biological annotation remains important. Curated biological-process and pathway gene sets provide interpretability, mechanistic context, and a structured vocabulary for organizing transcriptomic variation. Random gene sets lack this direct interpretative grounding. However, annotation alone does not guarantee clinical usefulness. A biologically named observable must still be tested for endpoint alignment, random-background advantage, and internal coherence. In this framework, biological structure is not invalidated by random high- D results; it is made more accountable.

The key conclusion is that high-discriminability random observables do not undermine the use of structured biological-process observables. Instead, they reveal the need for a multi-layer interpretation strategy. Discriminability identifies endpoint-aligned separation. Random-background comparison evaluates whether that separation exceeds what may occur by random aggregation.

Coherence evaluates whether the observable is internally compact in activity–width space. Biological-overlap analysis evaluates whether apparently random signals may contain partial structured biology.

Thus, random high- D results are not treated as contradictions. They are treated as controls, warnings, and potential clues. They prevent overinterpretation of survival-based D , clarify the distinction between clinical separation and biological meaning, and support the broader conclusion that process-level transcriptomic observables should be interpreted through activity, width, random-background behavior, and biological structure together.

Supplementary Note S5

Observable geometry, scale sensitivity, and uncertainty-aware discriminability

The activity–width representation used in this study treats each biological-process or pathway-level observable as more than a scalar activity score. Instead, each observable is characterized by both its activity magnitude and its internal width. This creates a minimal observable geometry in which a process-level feature can be described by where it is located in activity space and how broad or compact its internal gene-expression distribution appears. Under partial observability, this geometry is useful because the same mean activity may correspond to different internal observable structures.

In conventional pathway-level analysis, a biological process is often summarized by a single score. Such a score is useful, but it compresses the internal distribution of member-gene expression values into one number. The present framework separates two components of this compressed representation. The mean activity $\mu_{B,p}$ describes the central tendency of biological-process observable B in patient p , whereas the patient-level observable width $W_{B,p}$ describes the internal dispersion of the same observable. Together, $\mu_{B,p}$ and

$W_{B,p}$ form a simple activity–width coordinate system for process-level transcriptomic observation.

This activity–width geometry is not intended to reconstruct the full latent cancer state. Rather, it provides an operational description of how process-level transcriptomic observables behave under projected bulk measurement. A compact observable may indicate that genes within a process behave relatively coherently in a given patient, whereas a broad observable may indicate heterogeneous gene-level behavior, mixed subprogram activation, biological crosstalk, cellular composition effects, or measurement variability. Therefore, width modifies the interpretation of activity. A high activity score with narrow width and a high activity score with broad width should not necessarily be treated as equivalent observations.

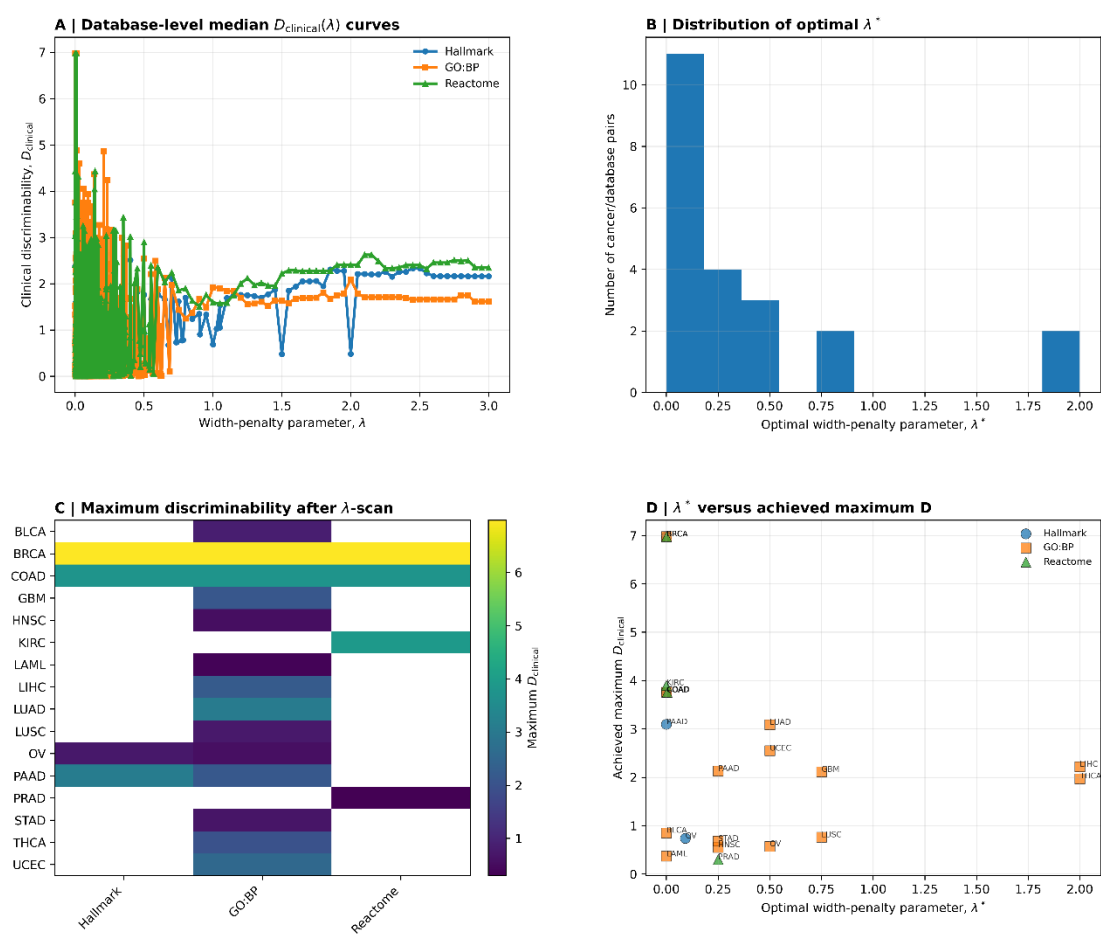
This is the motivation for uncertainty-aware discriminability. Survival-based clinical discriminability evaluates whether an observable separates patients into groups with different outcomes. However, a discriminative observable may still have broad internal width. In such a case, the observable may separate outcomes but remain internally heterogeneous. The uncertainty-aware formulation therefore introduces width as a diagnostic penalty or interpretative modifier, not as a claim that width universally improves outcome separation.

The λ -scan analysis was included to examine this issue. In the uncertainty-aware score, λ controls the relative contribution of the width penalty. When λ is small, the score is driven mainly by activity magnitude or endpoint-based separation. When λ is larger, broader observables are penalized more strongly. The λ -scan therefore evaluates how sensitive candidate observable selection and discriminability are to assumptions about the importance of observable width.

λ should not be interpreted as a universal biological constant. It is an analysis parameter used to test whether width-aware interpretation is stable across different penalty strengths. If the same or similar observables remain informative

across a broad range of λ values, this suggests that their usefulness is not highly dependent on a narrow parameter choice. If the selected observables change substantially as λ changes, this indicates that the ranking is sensitive to how strongly observable dispersion is penalized. Such sensitivity is itself informative, because it reveals that clinical discriminability and observable compactness may not always align.

Supplementary Figure S2. Full lambda-scan analysis across cancers and pathway databases



λ controls the width penalty and is interpreted as a sensitivity parameter, not a universal biological constant.

Supplementary Figure S2. Full λ -scan analysis across cancers and pathway databases.

(A) Database-level median clinical discriminability curves across λ values. The parameter λ controls the strength of the width penalty in the uncertainty-aware discriminability score. The curves show that discriminability changes with λ and

differs across gene-set databases, supporting the interpretation of λ as a sensitivity parameter rather than a universal biological constant. (B) Distribution of optimal λ^* values across cancer/database pairs. Most optimal values were located in the low- λ range, while a small number of cases showed higher λ^* , indicating context-dependent sensitivity to width penalization. (C) Maximum clinical discriminability achieved after λ -scan for each cancer and pathway database. (D) Relationship between λ^* and achieved maximum clinical discriminability. These results indicate that high discriminability is not associated with a single fixed λ value and that width-aware scoring should be interpreted as a diagnostic sensitivity analysis rather than a universal optimization rule.

The cross-cancer λ -scan results are summarized in Supplementary Figure S2. These results show that the width-penalty parameter affects discriminability in a context-dependent manner and should therefore be interpreted as a sensitivity parameter rather than a universal biological constant.

Scale sensitivity enters the analysis in two distinct ways. The first concerns the internal size of a biological-process observable, denoted N_B . N_B is the number of mapped genes within a single biological-process or pathway observable B . It affects how $\mu_{B,p}$ and $W_{B,p}$ are estimated. Very small gene sets may produce unstable width estimates or be driven by a few genes, whereas very large gene sets may combine multiple subprograms and produce broader or diluted observable structure. Therefore, N_B reflects the internal aggregation scale of one process-level observable.

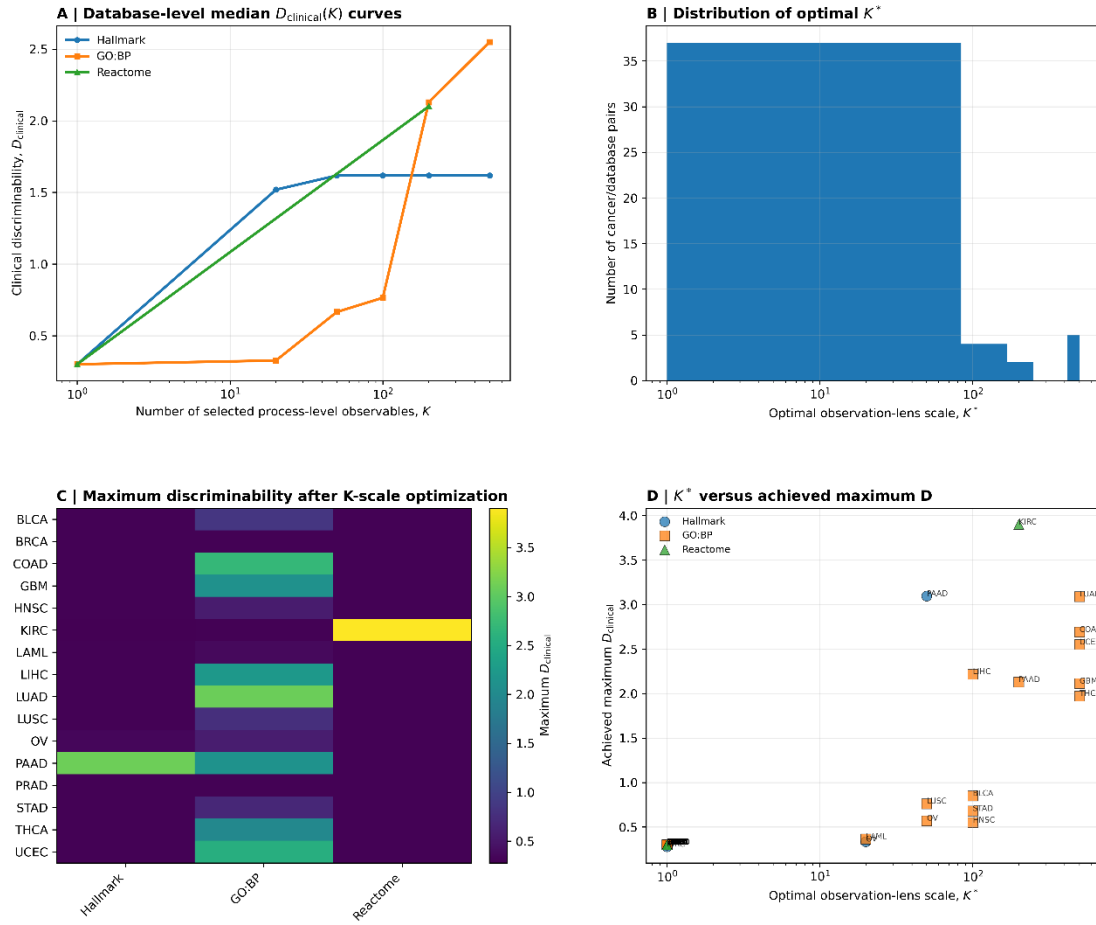
The second scale parameter is K . K denotes the number of selected process-level observables used to construct a patient-level score. This is conceptually different from N_B . Whereas N_B concerns the number of genes inside one observable, K concerns the number of observables included in a higher-level patient stratification score. Confusing N_B and K would mix two different levels of scale: gene aggregation within a biological process and observable selection across biological processes.

The distinction between N_B and K is further illustrated in Supplementary Figure S1D, where N_B affects the width of individual process-level observables, whereas K -scale analysis evaluates the number of selected observables used for patient-level scoring.

This distinction is central to the interpretation of the K -scale observation-lens analysis. Increasing K does not simply mean adding more genes. It means adding more process-level observables to the patient-level representation. If additional observables are informative and non-redundant, increasing K may improve clinical discriminability. However, if additional observables are weakly aligned with the endpoint, redundant, noisy, or internally broad, increasing K may reduce or fail to improve discriminability. Therefore, more observables do not automatically produce a better clinical representation.

The K -scale analysis evaluates whether there is an effective observation-lens scale for patient stratification. A small K may capture only a narrow part of the outcome-relevant biology. A very large K may introduce noise, redundancy, or weakly relevant processes. An intermediate K may sometimes provide better separation if it balances signal inclusion and noise control. This does not imply that a universal optimal K exists across all cancer types or endpoints. Rather, it indicates that the useful scale of process-level observation is context-dependent.

Supplementary Figure S3. Full K -scale observation-lens analysis across cancers and pathway databases



K denotes the number of selected BP/pathway observables in a patient-level score; it is distinct from N_B , the number of genes within one observable.

Supplementary Figure S3. Full K -scale observation-lens analysis across cancers and pathway databases.

(A) Database-level median clinical discriminability curves across different values of K , where K denotes the number of selected process-level observables included in a patient-level score. The curves show that discriminability changes with observation-lens scale and differs across gene-set databases. (B) Distribution of optimal K^* values across cancer/database pairs. The broad distribution indicates that no single observation-lens scale is uniformly optimal across all cancer contexts. (C) Maximum clinical discriminability achieved after K -scale optimization for each cancer and pathway database. (D) Relationship between K^* and achieved maximum clinical discriminability. These results support the interpretation that adding more process-level observables does not

necessarily produce a universal improvement in discriminability; instead, the useful observation scale is context-dependent. K is distinct from N_B , which denotes the number of mapped genes within one biological-process or pathway observable.

The full K -scale observation-lens analysis is summarized in Supplementary Figure S3. These results show that the optimal number of selected process-level observables varies across cancer/database pairs, supporting the interpretation that discriminability depends on observation scale rather than simply increasing with the number of included observables.

Together, λ and K evaluate different aspects of observable geometry. λ tests how strongly width should influence the interpretation of process-level activity and discriminability. K tests how many selected process-level observables should be included in a patient-level score. N_B , by contrast, describes the gene-level scope of each individual biological-process observable. These three quantities therefore represent different layers of scale and interpretation:

- N_B : genes within one observable
- K : number of selected observables
- λ : strength of width-aware penalty

This separation helps clarify the structure of the analysis. The study does not claim that discriminability is determined only by activity magnitude. Nor does it claim that broader width is always bad or that narrower width is always clinically superior. Instead, discriminability is treated as geometry- and scale-dependent. A useful observable should be evaluated according to its activity magnitude, internal width, survival-based endpoint alignment, random-background behavior, biological structure, and sensitivity to scale.

The integrated interpretation is therefore cautious. A high clinical D value identifies an observable or score that separates outcome groups under the chosen endpoint. A favorable activity–width profile suggests that the observable

may be internally coherent. A random-background advantage suggests that the observed discriminability is less likely to be explained by size-matched random aggregation alone. A stable λ response suggests that the result is not overly dependent on one width-penalty setting. A meaningful K -scale pattern suggests that the number of selected observables matters for patient-level representation.

This geometry-based view also explains why activity-only and width-aware scores may disagree. An activity-only score may prioritize observables with strong mean shifts, even when their internal expression distributions are broad. A width-aware score may down-rank such observables if their dispersion suggests weaker compactness or greater ambiguity. Conversely, some observables with moderate activity but compact width may become more favorable when width is considered. These differences do not indicate that one representation is universally correct. They show that activity and width capture complementary aspects of projected transcriptomic observation.

The main value of uncertainty-aware discriminability is therefore interpretative rather than absolute. It provides a way to ask whether an observable is not only endpoint-aligned but also compact under the chosen biological-process projection. This is particularly important in bulk transcriptomic data, where the same pathway score may arise from different combinations of cellular composition, subpathway activation, and gene-level heterogeneity. Width-aware interpretation helps expose this hidden structure instead of treating all equal mean scores as equivalent.

At the same time, the analysis has clear limits. λ -scan and K -scale optimization should not be interpreted as discovering universal biological constants or complete latent-state coordinates. They are diagnostic procedures for evaluating sensitivity, robustness, and observation scale under available data. Their results depend on the cancer cohort, gene-set database, endpoint, preprocessing choices, stratification method, and random-background design. Therefore, they should be used to support cautious interpretation rather than overstate generality.

In summary, observable geometry, scale sensitivity, and uncertainty-aware discriminability are connected by a common idea: process-level transcriptomic observables should not be judged by activity alone. Under partial observability, the usefulness of an observable depends on how its signal is organized, how broad or compact it appears, how it compares with random background, and how it behaves across observation scale. The activity–width representation, λ -scan, and K -scale analysis provide complementary diagnostic tools for evaluating these properties. Their purpose is not to replace biological validation, but to make the interpretation of pathway-level transcriptomic observables more transparent, cautious, and structurally informed.

Representative top-performing process-level observables are summarized in Supplementary Figure S5. These results show that high width-aware discriminability is distributed across cancer cohorts and gene-set databases, while the database contribution pattern should be interpreted descriptively because it depends on the size, annotation scope, and composition of each gene-set universe.

Supplementary Note S6

Boundaries and limitations of inference from projected transcriptomic observables

The present study interprets bulk transcriptomic measurements as projected observations of heterogeneous cancer systems. This perspective is useful because it makes explicit that gene-expression data are not direct readouts of a unique latent biological state. However, it also requires clear boundaries. The activity–width framework, survival-based discriminability analysis, random-background diagnostics, and K -scale observation-lens analysis are intended to support cautious interpretation of process-level observables. They should not be interpreted as complete latent-state reconstruction, causal mechanism identification, or universal clinical prediction.

First, the framework does not reconstruct the true latent state of the cancer system. Bulk gene-expression profiles are generated from mixed tumor tissues containing malignant cells, immune cells, stromal components, vascular structures, and other hidden biological sources. A single observed transcriptomic profile may therefore be compatible with multiple underlying cellular compositions, regulatory states, spatial arrangements, clonal configurations, or microenvironmental conditions. The process-level observables used in this study provide structured projections of this hidden system, but they do not uniquely identify it.

Second, the observable width $W_{B,p}$ should not be interpreted as an exact entropy or a full posterior uncertainty estimate. In this study, $W_{B,p}$ summarizes the within-process dispersion of z-scored gene-expression values for biological-process observable B in patient p . A broader width suggests greater internal dispersion within that process-level projection, while a narrower width suggests greater compactness. However, this width is an operational proxy. It does not directly measure the entropy of the latent cancer state, the entropy of the tumor microenvironment, or the full uncertainty of $P(S | O)$.

Third, survival-based discriminability is an operational endpoint-alignment measure, not a causal proof. A process-level observable with high survival-based D may capture clinically relevant variation, but it does not prove that the observable causes the survival difference. The association may reflect direct biology, indirect correlation with another biological axis, clinical confounding, tumor purity, treatment context, stage composition, immune infiltration, or other unmeasured factors. Similarly, a low- D observable is not necessarily biologically irrelevant. It may be important for tumor biology but poorly aligned with the survival endpoint, underpowered by event count, or masked by clinical heterogeneity.

Fourth, pathway-level and biological-process-level observables are coarse-

grained approximations. They are useful because they reduce high-dimensional gene-expression data into interpretable biological coordinates. However, averaging genes within a pathway can hide subpathway structure, gene-specific directionality, network topology, cell-type specificity, and regulatory hierarchy. The activity–width representation partly addresses this by pairing mean activity with internal width, but it does not fully recover the lost microscopic structure.

Fifth, random-background comparison is a diagnostic control, not a complete null model of biology. Size-matched random gene sets help evaluate whether structured biological-process observables exceed what may occur under random aggregation. However, random gene sets drawn from the transcriptome are not biologically empty in a strict sense. Because cancer transcriptomes contain broad correlated programs, random gene sets may accidentally capture tumor purity, proliferation, immune content, stromal activity, or other outcome-relevant axes. Therefore, high- D random sets should be interpreted as warnings and diagnostic clues, not as proof that curated biological structure is irrelevant.

Sixth, coherence and biological-overlap checks improve interpretability but do not prove mechanism. A high- D observable that also shows favorable coherence and exceeds random-background thresholds is a stronger candidate for interpretation than a high- D observable alone. However, this still does not establish causal mechanism. Biological overlap with known annotations can suggest interpretability, but it cannot replace experimental perturbation, single-cell validation, spatial profiling, or longitudinal response data.

Seventh, λ -scan and K -scale analyses are sensitivity analyses rather than universal optimization procedures. The parameter λ controls the strength of the width-aware penalty in the uncertainty-aware score. It should not be interpreted as a fixed biological constant. Similarly, K represents the number of selected process-level observables used to construct a patient-level score. An apparent optimal K in one cohort or endpoint does not imply a universal observation scale for cancer. Both λ and K depend on the dataset, endpoint, gene-set universe,

preprocessing choices, stratification method, and random-background design.

Eighth, the results are limited by the nature of bulk transcriptomic cohorts. Public cancer datasets are valuable because they provide relatively large sample sizes and clinical annotations, but they are usually cross-sectional. They often lack matched spatial information, longitudinal sampling, treatment timing, treatment response, clonal evolution, tumor purity estimates, and detailed microenvironmental composition. Therefore, the present analysis can evaluate endpoint-aligned observable structure, but it cannot directly track state transitions, treatment-induced movement, or causal progression trajectories.

Ninth, the framework should not be interpreted as a universal clinical decision model. The process-level observables identified in this study are not proposed as ready-to-use clinical biomarkers. Their usefulness depends on external validation, endpoint specificity, cohort reproducibility, assay platform compatibility, and clinical context. A candidate observable that performs well in one dataset may fail in another due to differences in patient population, treatment regimen, sample processing, follow-up duration, censoring structure, or event frequency.

Tenth, the study does not claim that width-aware scoring is always superior to activity-only scoring. The central claim is more modest. Observable width changes how discriminability should be interpreted. In some cases, width-aware analysis may improve candidate selection or reveal hidden heterogeneity. In other cases, activity-only scores may perform similarly or better for a specific endpoint. The purpose of $W_{B,p}$ is therefore not to replace conventional pathway activity scores, but to provide an additional diagnostic dimension for interpreting them under partial observability.

These limitations define the proper scope of the study. The framework is best understood as a minimal operational approach for evaluating projected transcriptomic observables. It asks whether process-level activity, internal width, random-background behavior, coherence, and observation scale provide

additional interpretative structure when assessing endpoint-aligned discriminability. It does not claim to infer the complete hidden cancer state, identify definitive causal mechanisms, or provide universal clinical rules.

The appropriate conclusion is therefore cautious. Bulk transcriptomic observables can be useful, but their interpretation should account for projection, heterogeneity, uncertainty, random-background behavior, and scale dependence. The activity–width representation provides one practical way to expose part of this structure using available gene-expression data. Future validation using longitudinal, single-cell, spatial, perturbational, and treatment-response datasets will be necessary to determine how closely these projected observables approximate underlying biological state transitions and clinically actionable mechanisms.

Supplementary Note S7

Future directions

The present study provides a minimal operational framework for interpreting process-level transcriptomic observables using activity, observable width, random-background behavior, coherence, and scale sensitivity. Several future directions may extend this framework, but these extensions should be viewed as next-step validation tasks rather than conclusions of the current study.

First, future work should evaluate whether activity–width representations are reproducible across independent cohorts and platforms. The present analysis focuses on publicly available bulk transcriptomic cohorts, but stronger validation will require external datasets, independent cancer cohorts, different assay platforms, and clinically matched endpoints. Such validation would help determine whether high-discriminability and width-aware observables are stable beyond the discovery datasets.

Second, the framework can be extended to endpoints that are closer to dynamic cancer-state change. Survival is clinically meaningful and widely available, but it is a coarse endpoint. Treatment response, recurrence, progression-free interval, pathological complete response, minimal residual disease, longitudinal molecular change, and pre/post-treatment transition would provide more direct tests of whether process-level observables capture clinically relevant state movement. Such endpoints may be especially useful for evaluating whether activity-width structure reflects progression, resistance, or therapeutic sensitivity.

Third, future studies should integrate additional molecular and tissue-level modalities. Bulk gene expression provides an effective projected observation, but it cannot fully resolve cellular composition, spatial organization, clonal structure, protein activity, epigenetic regulation, or signaling dynamics. Single-cell transcriptomics, spatial transcriptomics, proteomics, phosphoproteomics, copy-number profiles, mutation data, and histopathology could help determine whether broad or narrow observable width corresponds to identifiable biological sources of heterogeneity.

Fourth, network-aware extensions may improve interpretation of process-level observables. In the present study, gene sets are treated mainly as annotated biological-process units. However, genes within a process are not independent; they are embedded in regulatory, signaling, metabolic, and protein-interaction networks. Future work may examine whether observable width, coherence, and discriminability are shaped by network topology, module structure, pathway crosstalk, or proximity to cancer-relevant network regions.

Fifth, future analysis may refine the uncertainty-aware score. In this study, width is used as a simple operational proxy for observable dispersion, and λ is treated as a sensitivity parameter rather than a biological constant. Future studies could explore alternative uncertainty measures, adaptive λ selection, endpoint-specific weighting, Bayesian formulations, or model-based links between observable width and posterior uncertainty. These extensions should be validated carefully to

avoid overfitting or overinterpreting projected measurements.

Finally, the activity–width framework may be useful for constructing more cautious biomarker-selection workflows. Instead of selecting candidate observables solely by pathway name or survival separation, future workflows could jointly evaluate endpoint discriminability, random-background advantage, observable coherence, reproducibility, and biological interpretability. This may help identify process-level observables that are not only clinically associated, but also structurally interpretable under partial observability.

Overall, these future directions point toward a more comprehensive understanding of projected molecular observables in cancer systems. The current study provides only a minimal starting point: it shows that process-level transcriptomic observables can be interpreted through activity and width, and that discriminability should be evaluated together with uncertainty, random background, biological structure, and observation scale. Further experimental, longitudinal, spatial, and multi-omics validation will be needed before these ideas can support stronger biological or clinical claims.